Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/04/2012
PI Name:	Dickman, J. David Ph.D.		
Project Title:	Altered Gravity Effects Upon Vestibular Adaptation, Development and Regulation of Bone Homeostasis		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	ddickman@cns.bcm.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-798-4589
Organization Name:	Baylor University/Baylor College of Medicine		
PI Address 1:	Department of Neuroscience		
PI Address 2:	One Baylor Plaza, MS: 295		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030-3411	**Congressional District:**	7
Comments:	NOTE PI moved to Baylor in Fall 2011; previously at Washington University, St. Louis, MO (Ed., 12/20/2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	02/01/2010	**End Date:**	01/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ornitz, David (Washington University)		
Grant/Contract No.:	NNX10AE38G		
Performance Goal No.:			
Performance Goal Text:			

Exposure to microgravity or hypergravity environments provides new adaptive challenges for many of the body's major systems. The primary objective of the current collaborative project was to examine the effects of exposure to altered gravity conditions upon vestibular receptor morphology and regulatory mechanisms of bone homeostasis. A number of established and recent investigations have shown that exposure to altered gravity conditions can significantly affects all of these processes. Examining the effects of altered gravity conditions upon vestibular receptor formation is possible using normal animals placed in either hypergravity or appropriately controlled rotational 1g environments. Yet, investigations regarding vestibular regulatory functions are multivariate and difficult to interpret. However, the recent development of transgenic mice with phenotypes that either lack the ability to sense gravity conditions or are completely normal has provided new investigative tools. Here, heterozygous Otopetrin mice (Otop1) that harbor a ßgal reporter

	gene and exhibit a normal otoconia phenotype, as well as homozygous Otopetrin deficient mice (Otop1-) exhibit a complete absence of vestibular receptor otoconia were used. Otop1 mice were hypothesized to sense changes in gravity load that result in altered vestibular function and an increase in vestibular regulation of bone remodeling when exposed to long-duration hypergravity (2g). In contrast, Otop1- mice, where no vestibular response to changes in gravity load are detectable, were hypothesized to exhibit no relative change due to altered gravity exposure as compared to heterozygous controls. In fact, use of the heterozygous and homozygous Otopetrin mice offers an ideal control for examining vestibular regulatory mechanisms in response to changes in gravity conditions, since other effecting variables, such as stress and skeletal load, will be identical for the two groups of mice.
Task Description:	Here, we first finished experiments in gene expression in quail, looking at otoconia development during embryogenesis and the genes that regulate their growth. We found moderate changes in gene expression for Otop 1 and OC90 at embryonic day E9 in quail, although the overall gene expression patterns were not different for animals raised in normal 1g and hypergravity 2g conditions. We then switched to mice and began early embryonic exposure of litters to hypergravity conditions. None of these animals survived, as the mothers killed the pups, likely due to environmental stresses. We then adopted a different strategy by breeding 4 pairs of mice in 1G rotation (center of centrifuge) and hypergravity 2g habitats. We had two successful breeding cycles where the pups were nurtured until 4 weeks In the proposed studies, when we harvested the animals and tissue samples of inner ear receptor and limb bones were obtained. We are continuing the analyses of these organs to examine the effects of altered gravity exposure upon vestibular receptors and bone formation. We are examining the effects of altered gravity upon vestibular receptor formation by examining receptor cells, afferent nerve innervation, Otopetrin expression, and otoconia synthesis. We will investigate the effects upon vestibular regulation of bone homeostasis by examining bone density measures of the forelimb and hind limbs using micro computerized tomography. These studies promise to provide crucial information into the role of gravity as a determinant for vestibular system development, adaptation, and regulation of other major systems. Before long-term space missions are manifested for habitation on other planets, moons, or environments where altered gravity conditions are certain, it is crucial to understand how the vestibular system will adapt in structure, function, and regulatory behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of the vestibular system that coordinates all of our movement, stabilizes our gaze, and provides directional information for spatial orientation is highly dependent upon the presence of gravity. In fact, gravity has been shown to be powerful determinant for development in many systems, including the nervous, muscular, and skeletal systems. Our research will directly assess how these systems utilize information derived from vestibular receptors as epigenetic determinants for normal system development, as well as adaptation to altered gravity environments. When vestibular receptors are compromised, what are the consequences to normal development? Our work will provide insights regarding vestibular receptors and bone formation in normal animals and those exposed to altered gravity conditions, such as may be encountered during space habitation, or growth on extraterrestrial bodies.
Task Progress:	Previous investigations found that altered gravity exposures affect inner ear receptor morphology and function, since these organs detect the orientation and intensity of gravity. Here, we examined receptor formation in the otolith organs in response to altered gravity conditions in quail and mice. The gravity detector particles, otoconia that are made of small calcium carbonate crystals, were examined to determine if altered gravity affects their formation. In quail, we examined the the weight of the otoconia fused stones and calcium content at two time points, embryonic days E12 and E14, when formation is maturing. No overall average difference due to gravity was observed. However, when examined individual otoconia particles, we found significant differences in the size of individual otoconia particles. We found that the lengths and widths of otoconia were smaller in 2G embryos. The ratios of the length to width decreased for utricular otoconia at both E12 and E14 2G embryos, but remained constant for the larger saccular otoconia. Together, our results indicate the hypergravity produces changes in otoconia size across the receptor epithelium, but does not significantly affect the overall fused stone weight, calcium content. We also examined the relative expression of two genes that regulate otoconia growth, Oc90 and Otop1 for all embryonic development days in quail. We found that Oc90 was minimal in the receptors early in development through E7, at values below 10% of that observed at the highest level which occurred at embryonic day E9. At E10, there was a sharp decline in Oc90 expression levels that continued to decrease toward hatching. For Otop1, the mean expression profiles were similar to those observed for Oc90. The expression level during early development increased greatly at E6, coincident with the initial formation of immature otoconia. Maximal levels of Otop1 were observed at E9. After E9, the Otop1 levels dropped dramatically at E10, further at E12, then remained low until hatching. Next, we examined the expression patterns for both Oc90 and Otop1 in 2G embryos during higher levels of gravity (2g) development. We observed that the highest level of expression for both Oc90 and Otop1 occurred at E9, similar to 1g embryos. However, there was a significant decrease in both Oc90 and Otop1 expression in the 2G embryos as compared to 1G animals for only the E9 time point. At all other developmental time points, no differences in expression were found. This work forms a portion of a paper that is currently under review.
Bibliography Type:	Description: (Last Updated: 08/06/2012)
Articles in Peer-reviewed Journals	Kim E, Huss D, Ignatova E, Thalmann R, Thalmann I, Ornitz D, Dickman JD. "Development of otoconia and expression of Otopetrin 1 and Otoconin 90 in quail raised in normal and hypergravity environments." Developmental Neurobiology. In revision, as of August 2012. , Aug-2012